

Mirza Md Tasnim Mukarram (He/Him)



906 Benton Drive, Apt 13, Iowa City, IA 52246 | mtasnimukarram@uiowa.edu
Cell: (319) 800-8098 | Website: github.com/Mirza9003
FAA 107 Certified Remote Pilot, United States of America

EDUCATION

University of Iowa PhD in Geographic Information Science & Cartography	Iowa City, USA · Aug 2024 – Present
University of Iowa Master's in Informatics	Iowa City, USA · Aug 2025 – Aug 2027 (Expected)
United International University (UIU) PGD in Data Science	Dhaka, Bangladesh · March 2024
Dhaka School of Economics, University of Dhaka Master's in Environmental Economics	Dhaka, Bangladesh · July 2023
Military Institute of Science & Technology (MIST) BSc in Civil & Environmental Engineering	Dhaka, Bangladesh · January 2019

EXPERIENCE

Graduate Research Assistant (GRA) Iowa Technology Institute, College of Engineering — NSF EPSCoR RII FEC (DAREWICCAR)	Iowa City, USA · June 2025 – Present
- Conduct advanced machine learning modeling to forecast extreme weather events (floods, droughts, heatwaves) in the KANI region (Kansas, Arkansas, Nebraska, Iowa). - Analyze high-resolution climate and remote sensing data to assess localized climate change impacts on agricultural resilience. - Develop predictive frameworks integrating environmental datasets to improve regional weather intelligence systems. - Design and deliver hands-on ML training workshops for undergraduate students on climate and environmental data science. - Collaborate with interdisciplinary teams to support NSF-funded research objectives and contribute to publications and technical reports.	
Graduate Teaching / Research Assistant (GTA/GRA) GIS Lab at School of Earth, Environment & Sustainability	Iowa City, USA · Aug 2024 – May 2025
Facilitate student learning, assessment etc..	
Associate Environmental Engineer Bangladesh Army — HQ 24 Engineer Construction Brigade (Further Development of Comilla University Project)	Dhaka, Bangladesh · Feb 2021 – Aug 2024
- Mapped land cover, natural assets, fragile urban systems, climate risk, and vulnerable hotspots. - Prepared city profiles, vulnerability assessments, and greenhouse gas emission reports. - Developed project reports and Terms of Reference (ToR); identified climate change and environmental protection projects. - Managed project monitoring, produced high-quality military maps, and built and quality-controlled remote sensing databases.	
GIS & Water Quality Specialist Climate Change Laboratory, Green Laboratory Testing Services — BEPZA	Dhaka, Bangladesh · Jan 2020 – Jan 2021
- Monitored and tested water and soil quality parameters; conducted field visits and surveys. - Prepared confidential water quality maps and project reports; performed land suitability analysis. - Acquired data from satellite and ground-based platforms and prepared project completion reports.	
Graduate Research Assistant (GRA) Dept. of Civil Engineering, Military Institute of Science & Technology	Dhaka, Bangladesh · Jan 2019 – Dec 2019
- Performed land suitability analysis, GIS mapping, and statistical data analysis. - Prepared land use maps using RapidEye and Landsat imagery; conducted laboratory experiments and EIA consultancy. - Carried out field visits (FGD, KII, PI) and prepared presentations for seminars and conferences.	

AWARDS & ACHIEVEMENTS

- NSF I-GUIDE Travel Award for the Summer School 2026: Spatial AI & Convergence Science (Urbana Champaign, US)
- AAAS Travel Award for the 2026 EPSCoR Summit (New Orleans, LA, USA).
- Graduate Student Research Award 2025, Martha L. Corry Fund for Geography (University of Iowa).
- Best Research Project Award 2018 among all engineering departments at MIST.
- Full-fund scholarship (Korean Water Forum) to attend the 5th Asia Pacific Youth Parliament for Water 2016, DGIST, South Korea.
- Full-fund scholarship to attend the 9th Session of the World Urban Forum as a Delegate, 2017.

- Cherag Roy Memorial Scholarship (Wildlife Conservation), Bangladesh Python Project 2016.
- 3rd Place in Bangladesh, International Public Speaking Competition 2016 (ESU of Bangladesh).
- 2nd Place, Project Showcasing — Environment Fest 2018 (BUP); 2nd Place, Nuclear Fest 2018 (MIST).

SKILLS

Technical: ArcGIS, Google Earth Engine, Python, SPSS, STATA, Microsoft Office, Photoshop, Video Editing.

Languages: English (Full professional proficiency, IELTS 7.5), Bangla (Native), Urdu & Hindi (Speaking).

PEER-REVIEWED PUBLICATIONS

1. **Mukarram, M.Md.T.**, Rukiya, Q.U., Linderman, M., Wang, J., Al Kafy, A.A. (2026). Vision Transformer-Based Daytime Land Surface Temperature Modelling and Explainable Green-Roof Cooling Scenario Assessment using Airborne LiDAR and Satellite Data. *Sustainable Cities and Society*. (With editor since 16 May 2026).
2. Rukiya, Q.U., Linderman, M., **Mukarram, M.Md.T.** (2026). Depth-Dependent Soil Moisture Drought Dynamics and Vertical Propagation in the State of Iowa. *Theoretical and Applied Climatology*. (Submitted 1 June 2026).
3. **Mukarram, M.Md.T.**, Linderman, M., Shokrana, S. (2026). Geospatial Artificial Intelligence-based Flood Susceptibility Mapping and Urban Flood Risk Assessment. *Environmental Impact Assessment Review*. (Under review since 28 April 2026).
4. **Mukarram, M.Md.T.**, Waleed, M., Rukiya, Q.U., Linderman, M., Demir, I. (2026). Prioritizing Agricultural Flood Mitigation: A GeoAI-Driven Assessment of Susceptibility, Crop Exposure, and Socioeconomic Vulnerability. *Int. J. of Flood Risk Management*. (Under review since 21 April 2026).
5. **Mukarram, M.Md.T.**, Rukiya, Q.U., Linderman, M. (2025). Harnessing Machine Learning, AI and Remote Sensing for Drought Severity Assessment & Prediction in the USA. *Geosystems and Geoenvironment*. (2nd Revision, Elsevier).
6. **Mukarram, M.Md.T.**, Rukiya, Q.U., Linderman, M. (2025). AI-Driven Prediction of Microclimate Dynamics: A 20-Year Comparative Multi-Model Assessment Across the U.S. Midwest (KANI Region). *Modeling Earth Systems and Environment*. (2nd Revision).
7. Al Kafy, A.A., Nowrin, M.T., Altuwaijri, H.A., **Mukarram, M.Md.T.** (2026). Advanced hybrid ensemble learning models for atmospheric humidity prediction. *Int. J. of Environmental Science & Technology*. <https://doi.org/10.1007/s13762-025-06926-z>
8. Miah, M.T., Fariha, J.N., Al Kafy, A., **Mukarram, M.Md.T.**, et al. (2025). Geostatistical assessment of spatial climate dynamics using mono-window ML for land cover and demographic shifts. *Theoretical and Applied Climatology*, 156(10), 544. <https://doi.org/10.1007/s00704-025-05750-2>
9. **Mukarram, M.Md.T.**, Kafy, A.A., et al. (2025). IoT-Enabled Smart Boat System for Integrated Water Resource Management. *Water Resources Management*. (1st Revision, Springer).
10. Kafy, A.A., ..., **Mukarram, M.Md.T.** (2025). Monitoring spatiotemporal changes in urban biophysical composition and LST dynamics in Saudi Arabia. *Environmental Monitoring and Assessment*. (Accepted with minor revisions).
11. Kafy, A.A., **Mukarram, M.Md.T.**, et al. (2025). Quantifying urban heat island dynamics in arid environments: a 20-year remote sensing analysis. *Geoscience Letters*. (Under peer review, Springer).
12. Kafy, A.A., **Mukarram, M.Md.T.**, et al. (2025). Two Decades of Urban Heat Island Evolution in Arid Cities: Al Taif, Saudi Arabia. (1st Revision).
13. **Mukarram, M.Md.T.**, Kafy, A.A., Rukiya, Q.U., Chowdhury, Md.A., Das (2023). Perception of coastal citizens on community-based rainwater harvesting for sustainable water resource management. *Resources, Conservation & Recycling* (IF 13.2). <https://doi.org/10.1016/j.resconrec.2023.107196>
14. Das, A., Shuvo, R.M., **Mukarram, M.Md.T.**, Sharno, M.A. (2024). Energy Policy Landscape in Developing Countries: A Stern Review and Meta-Analysis of Bangladesh. *Energy Strategy Reviews* (IF 8.2, Elsevier).
15. Khan, M.A.K., Ferdous, J., **Mukarram, M.Md.T.**, Nawshin, N., Rahman, M.T.U. Exploring safe water availability in southern coastal Bangladesh. (In preparation).

CONFERENCE PROCEEDINGS / PRESENTATIONS

1. Mukarram, M.M.T., Rukiya, Q.U., Linderman, M.A., Wang, J. (2026). GeoAI-Driven Climate Resilience using LiDAR Urban Morphology and Spatial Deep Learning. *AAAS Annual EPSCoR Summit 2026*, LA, USA.
2. Mukarram, M.M.T., Rukiya, Q.U., Linderman, M.A. (2025). Advancing Drought Prediction in the Midwest US using Remote Sensing, ML Models and Explainable AI (XAI). *AGU25*.
3. Shahrier, M., Al Kafy, A., Mukarram, M.M.T. (2025). Predicting Future Flood Vulnerability Zones in Texas Using Explainable GeoAI. *AGU25*.
4. Mukarram, M.M.T., Das, A. (2023). Probing the Potential of Green Markets Towards Sustainable Tourism Development of Bangladesh. *ICPACE, RUET, Bangladesh*.

5. Das, A., Mukarram, M.M.T. (2023). Co-Management of Localized Common Pool Natural Resource: A Sustainable Fisheries Management Regime in Bangladesh. ICPACE, RUET, Bangladesh.
6. Khan, M.A.K., Ferdous, J., Mukarram, M.M.T., et al. (2020). Spatio-temporal variation of groundwater quality in Mongla Upazila using IDW. ICEEST 2020, Dhaka.
7. Nawshin, N., Ferdous, J., Mukarram, M.M.T., Khan, M.A.K. (2020). Salinity Intrusion in Coastal Bangladesh: Change in Crop Production System. 4th ICSD 2020, Dhaka.
8. Mukarram, M.M.T., Khan, M.A.K., Ferdous, J., et al. (2020). Exorbitant shrimp cultivation inhibiting agro-based livelihoods in Mongla Upazila. 4th ICSD 2020, Dhaka.
9. Ferdous, J., Mukarram, M.M.T., Khan, M.A.K., Nawshin, N. (2020). Local warming triggered by wetland loss in Dhaka city. 4th ICSD 2020, Dhaka.
10. Khaled, A.M., Mukarram, M.M.T., et al. (2020). Sensor-Based Automated Rooftop Rainwater Collection System and Rainwater Quality Assessment of Dhaka City. ICACE-2020, CUET, Bangladesh.

SHORT COURSES

- Google Earth Engine · Statistical Data Analysis · STATA for Statistics and Data Analysis
- Urban Climate Change · Building Climate Resilient Migrant Friendly Cities (7th Annual ICCCAD, 5 days, Dhaka)
- Introductory E-course on Climate Change (1 month) · Causes of Climate Change (1 month)
- Mapping & Spatial Analysis with ArcGIS

WORKSHOPS & TRAININGS

- Zero Hunger, Zero Emissions University Workshop — ICCCAD / Environmental Change Institute, University of Oxford & Oxfam BD (5 days, Savar, Bangladesh).
- City Energy & Climate Action Plan Training — Action Asia (7 days, Kuala Lumpur, Malaysia).
- Youth Organizations Role & Youth Advocacy to Achieving the SDGs — Green Dialogue Series (1 day, Dhaka, Bangladesh).

SESSION GUEST SPEAKER & FACILITATOR

- Workshop speaker & facilitator, VNEC Earth Carnival 2018 — Viqarunnessa Earth Club, Dhaka.
- Workshop speaker & facilitator, Climate Change Resilience, Adaptation & Mitigation — Dept. of EWCE, MIST, Dhaka.
- Workshop speaker & facilitator, Introduction to Climate Change Science — Alpha Team Organization, BUP, Dhaka.

LEADERSHIP & VOLUNTARY ROLES

Climate Educate Project

Regional Manager, South Asia (Nov 2018 – June 2019) · Project Advocacy Head, Bangladesh (Oct 2016 – Oct 2018)

- Conducted climate change education workshops, training, and seminars in schools; prepared education documents; led project planning and communication.

Climathon — A Global Climate Action Program (2019)

Climate Mentor, Gen Lab — Program Management

- Trained participants and managed events.

World Water Congress & Exhibition

Paper Abstract Reviewer, International Water Association (IWA)

- Reviewed conference abstracts.

PROFESSIONAL AFFILIATIONS & PEER-REVIEW ROLE

- Institute of Engineers Bangladesh (IEB), Member A-23821.
- Asian Cities Climate Change Resilience Network (ACCCRN).
- Global Youth Climate Network.
- Journal Peer Reviewer (10+ manuscripts across multiple journals).

IN THE NEWS

- [PhD student Mirza Md Tasnim Mukarram — Travel Grant Award](#)

- [Invited to I-GUIDE Summer 2026, University of Illinois Urbana](#)
- [New article announcement — Advanced Hybrid Ensemble models](#)
- [New paper — publishes in Theoretical and Applied Climatology](#)